#Many Daughters

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Pre-Analysis Plan

Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes).

Outcome variable

We looked for survey questions that directly relate to gender equality. We did not find a question that would have been asked continuously over the years. Instead, we found three questions that capture attitudes towards gender equality. Unfortunately, each of these questions was only asked in a single year.

pl/plh0308_v2: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally"

- Only available in 2018
- Recoded (dichotomized):
    - recode plh0308_v2 (min/0 = .) (1/5 = 0) (6/7 = 1)
    - "1" is "agree"

pl/plh0308_v1: "The best scenario is when both the man and the woman are equally employed and both take equal responsibility for household and family care."

- Only available in 2012
- Recoded (dichotomized):
    - recode plh0308_v1 (min/0 = .) (1 = 1) (2/4 = 0)
    - "1" is "agree"

pl/plh0395i05: "Schools and preschools should promote non-gender-typical activities that are not typical for children's gender"

- Only available in 2019
- Recoded (dichotomized):
    - recode plh0395i05 (min/0 = .) (1/5 = 0) (6/7 = 1)
    - 1" is "agree"

We combine the three variables from the three years into one dichotomous variable representing support for gender equality (called "sge"):

```
egen sge = rowtotal(plh0308_v1 plh0308_v2 plh0395i05), missing
```

Treatment variable: Number of daughters ("ndaughters")

We use information contained in the biobirth dataset to calculate the number of daughters a respondent has had until the respective survey year. For each child, we observe the year it was born

(kidgeb*) and its sex (kidsex*). ndaughters is zero if the respondent has had no children by the year of survey and equals the number of kids that are recorded as female and born before or in the respective survey year.

Control variables

pgen/pgfamstd: Marital Status

- Recode: combine some categories
    - Recode pgfamstd (min/0 = .) (1 6 7= 1 “married”) (3 = 2 “single”) (2 4 8 = 3 “separated/divorced”) (5 = 4 “widowed”)

ppathl/sex: gender

hgen/hghinc: monthly net income of the household

- Recode: indicate missing values
    - Recode hghinc (-3/-1 = .)

pgen/pgisced97: Level of education

- Recode: combine some categories
    - Recode pgisced11 (min/-1 = .) (0/2 = 1 “less than secondary”) (3/4 = 2 “secondary”) (5/6 = 3 “higher”)

pgen/pglfs: Labor Force Status

- Recode: combine some categories
    - Recode pglfs (min/0 = .) (11 12 = 1 “employed”) (1/5 8/10 13 = 2 “not employed”) (6 = 3 “unemployed”)

Derived variables

nchildren: Number of children

- number of children born to the respondent before the year of survey(using kidgeb* in biobirth)

age and age_sq: Age and squared age

- age = survey year – year of birth (ppathl/gebjahr)
- age_sq = age^2

hhsize: Household size

- Number of observations in the same household in a given year

Inclusion/exclusion criteria:

We keep only observations in private households (pop == 1 | pop==2).

Due to the limited amount of data we found to test H1, we are restricted to the survey years 2012, 2018, and 2019. We will include all observations with non-missing values in all variables.

Exact analysis:

We follow the methodology of Oswald and Powdthavee (2010), but treat the outcome variable as cardinal and estimate the following linear probability model:

```
reg sge ndaughters i.nchildren i.pgfamstd i.sex age age_sq hghinc
hhsize i.pgisced11 i.pglfs i.syear
```

Interpretation of results:

We examine the results of the `reg` command. We consider the hypothesis to be empirically supported if the estimated coefficient on the number of daughters is positive and the p-value is less than 0.1.

Hypothesis 2: Having daughters leads to more left-leaning political preferences.

Variables

Outcome variable

pl/plh0012_h: Which party do you lean toward?

- available every year
- We choose the following party classification:
  - Left: SPD, Buendnis90/Gruene, Die Linke (and unambiguous combinations thereof)
  - Right: CDU, CSU, FDP, NPD/Republikaner/Die Rechte, AfD (and unambiguous combinations thereof)
- recode as dichotomous variable (left-leaning preference – "left")
  - recode plh0012_h (-8/-1 = .) (1 5 6 9 16 17 = 1 "left") (2 3 4 7 13 14 21 22 25 27 30 = 0 "right") (else = .), gen(left)

We use the same treatment variables and control variables as for Hypothesis 1.

Inclusion/exclusion criteria

We keep only observations in private households (pop == 1 | pop==2).

We will include all years and all observations with non-missing values in all variables.

Exact analysis

We follow the methodology of Oswald and Powdthavee (2010), but treat the outcome variable as cardinal and estimate the following linear probability model regression (with individual fixed-effects):

```
xtset pid syear
```

```
xtreg left ndaughters i.nchildren i.pgfamstd i.sex age age_sq hghinc
hhsize i.pgisced11 i.pglfs i.syear, fe
```

Interpretation of results

We examine the results of the `xtreg` command. We consider the hypothesis to be empirically supported if the estimated coefficient of the number of daughters is positive and the p-value is less than 0.1.

Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables

Outcome variable

We construct a measure of gender equality based on the differences between male and female partners' labor market participation and involvement in domestic chores.

pgen/pgtatzeit: Actual Work Time Per Week

- recode pgtatzeit (-2 = 0) (-1 -3 -7 = .)
- not working is treated as zero hours

pgen/ pglabgro: Current Gross Labor Income in Euro

- recode pglabgro (-2 = 0) (-5 -7 = .)
- not working is treated as zero labor income

Time-Use Variables

pl/pli0043_h: Hours Weekday Doing Housework (cooking, cleaning)

pl/pli0040: Hours Weekday Running Errands

pl/pli0044_h: Hours Weekday Childcare

pl/pli0046: Hours Weekdays Care For Persons

pl/pli0049_h: Hours Weekdays Performing building flat / apartment / house repairs, car repairs, gardening work

- recode all time-use variables such that negative values are set to missing
- generate domestic time-use variable:

```
egen domtime = rowtotal(pli0043_h pli0040 pli0044_h pli0046
pli0049_h), missing
```

We identify the respondent's partner via the variable pgen/pgpartnr and merge that partner's information on work time, labor income and time use to the respective respondent. The female partner's data is stored as outcome variable x_female and the male partner's as x_male, where x is either wt (worktime), ly (labor income) or domtime (time spent on domestic chores).

We then calculate an equality measure for each outcome variable $\boldsymbol{x}$ by

$$\boldsymbol{E_x = 1 - 2\left|\frac{x_female}{x_male \ + x_female} - \frac{1}{2}\right|}$$

We calculate a composite gender equality measure ("geneq") as the arithmetic mean of the standardized values (z_x) of each equality measure.:

```
egen geneq = rowmean(z_wt z_ly z_domtime)
```

We use the same treatment variables and control variables as for Hypothesis 1.

Inclusion/exclusion criteria

We keep only observations in private households (pop == 1 | pop==2). We restrict the sample to individuals living in heterosexual married or cohabiting couples (based on variable pgpartnr) with complete information on their own and their partner's labor market participation and time use. We will include all years and all observations with non-missing values in all variables.

Exact analysis

We follow the methodology of Oswald and Powdthavee (2010), but treat the outcome variable as cardinal and estimate an OLS regression (with individual fixed-effects) of the composite gender equality measure geneq:

```
xtset pid syear
```

```
xtreg geneq ndaughters i.nchildren i.pgfamstd i.sex age age_sq
hghinc hhsize i.pgisced11 i.pglfs i.syear, fe
```

Interpretation of results

We examine the regression results. We consider the hypothesis to be empirically supported if the estimated coefficient on the variable ndaughters is positive and the p-value is less than 0.1.

Hypothesis 4: ***Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)***

Variables

The variables (outcomes, treatment, and controls) are the same as for hypothesis 1.

Inclusion/exclusion criteria

Same as for hypothesis 1.

Exact analysis

We follow the methodology of Oswald and Powdthavee (2010), but treat the outcome variable as cardinal and estimate the following linear probability model (fully interacted with sex):

```
reg sge c.ndaughters##i.sex i.nchildren##i.sex i.pgfamstd##i.sex
c.age##i.sex c.age_sq##i.sex c.hghinc##i.sex c.hhsize##i.sex
i.pgisced97##i.sex i.pglfs##i.sex i.syear##i.sex
```

We calculate the marginal effects of the number of daughters on the probability of reporting strong support for gender equality for men and women separately and determine their difference:

```
margins, dydx(ndaughters) over(sex) post

lincom 1.sex-2.sex
```

Interpretation of results

We examine the regression results. We consider the hypothesis to be empirically supported if the estimated differences between the marginal effects of the number of daughters for fathers and mothers is positive and the p-value is less than 0.1.